

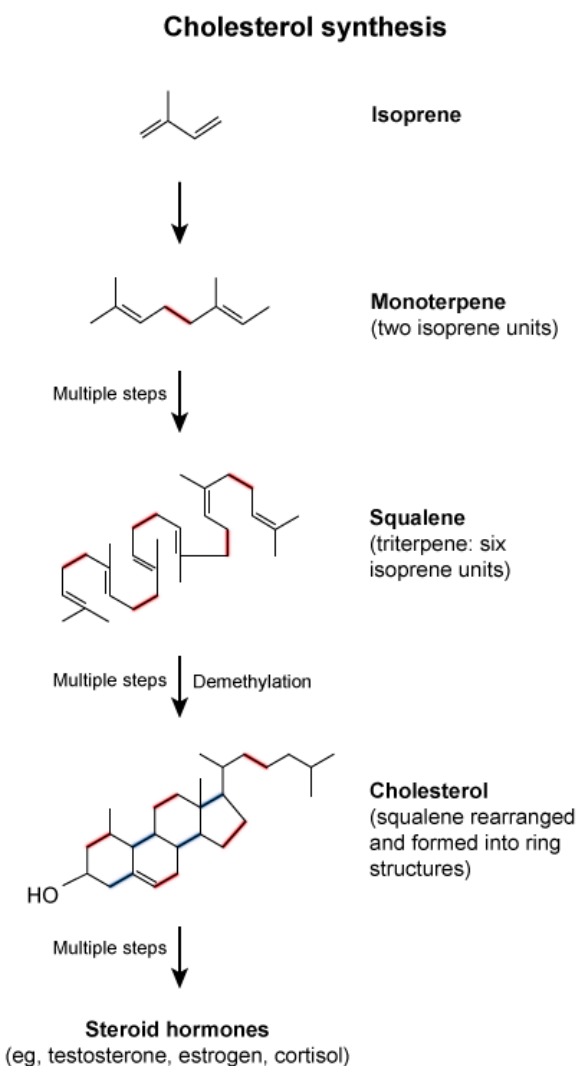
Which series depicts the order in which the precursors of steroid hormones are synthesized?

- ✔ ☒ A. Isoprene → monoterpene → squalene → cholesterol [52%]
☐ B. Cholesterol → isoprene → monoterpene → squalene [10%]
☐ C. Monoterpene → isoprene → squalene → cholesterol [35%]
☐ D. Isoprene → cholesterol → monoterpene → squalene [1%]

Correct

51% answered correctly

Explanation:



Note: Red bonds indicate links between isoprene units; blue bonds indicate locations of ring formation.
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Steroid hormones are involved in multiple critical signaling pathways in the body, including those related to physical development, reproduction, and stress. All **steroid hormones** are **synthesized from cholesterol** and share a **common backbone structure** consisting of four fused rings: 3 six-member rings and 1 five-

member ring.

The **basic unit of cholesterol** is a branched five-carbon molecule belonging to a class known as **isoprenes**. Isoprenes can combine with each other to form larger units known as **terpenes**. Terpenes consisting of two isoprene units are called **monoterpenes**, those consisting of two monoterpenes (or four isoprenes) are called **diterpenes**, and those consisting of three monoterpenes (six isoprenes) are called **triterpenes**. During cholesterol synthesis, a triterpene known as **squalene** forms. Squalene undergoes multiple steps, including demethylation, methyl rearrangements, and cyclization, to form cholesterol. The correct order of synthesis for the precursors of steroid hormones is **isoprene → monoterpene → squalene → cholesterol**.

(Choices B, C, and D) Each of these options either has something other than cholesterol as the final product or something other than isoprene as the starting material. Steroid hormones are synthesized from cholesterol, which is made of several isoprene units linked together.

Educational objective:

Steroid hormones are derived from cholesterol. Cholesterol has a characteristic four-ring backbone, which is synthesized from five-carbon subunits called isoprenes. Two isoprenes joined together form a monoterpene, and six isoprenes can join to form a triterpene called squalene, which then cyclizes and, after several steps, forms cholesterol.

Time spent:0 sec

Subject:

Biochemistry

QID:402028

System:

Carbohydrates, Nucleotides, and Lipids